

Entropy Bound for Empirical Non-Overlapping Blocks

Setup and notation (everything is deterministic)

Fix a finite alphabet $\mathcal{A} = \{1, \dots, K\}$ and a finite text

$$x_1 x_2 \cdots x_N \in \mathcal{A}^N,$$

with N a multiple of a chosen block length $n \geq 1$. Write $N = mn$ and split the text into *non-overlapping* blocks

$$B_s := (x_{(s-1)n+1}, \dots, x_{sn}) \in \mathcal{A}^n, \quad s = 1, \dots, m.$$

All objects below are defined by counting occurrences in this fixed text; there is *no* probabilistic model.

Empirical symbol distribution. For $a \in \mathcal{A}$,

$$p(a) := \frac{1}{N} \sum_{t=1}^N \mathbf{1}\{x_t = a\}, \quad p \in \Delta^{K-1}.$$

Empirical block distribution (non-overlapping). For $w \in \mathcal{A}^n$,

$$P(w) := \frac{1}{m} \sum_{s=1}^m \mathbf{1}\{B_s = w\}, \quad P \in \Delta^{K^n-1}.$$

Position- t marginals of blocks. For each $t \in \{1, \dots, n\}$ and $a \in \mathcal{A}$,

$$q_t(a) := \frac{1}{m} \sum_{s=1}^m \mathbf{1}\{x_{(s-1)n+t} = a\}, \quad q_t \in \Delta^{K-1}.$$

These are the marginals of P on coordinate t (with the non-overlapping segmentation).

Counting identity. For every $a \in \mathcal{A}$,

$$p(a) = \frac{1}{n} \sum_{t=1}^n q_t(a) \iff \boxed{p = \frac{1}{n} \sum_{t=1}^n q_t}. \quad (1)$$

Entropies

For a distribution r on a finite set, let $H(r) := -\sum r \log_2 r$ (with the convention $0 \log 0 = 0$). We write

$$H(p) \text{ for symbols,} \quad H(P) \text{ for } n\text{-blocks,} \quad H(q_t) \text{ for position-}t \text{ marginals.}$$

Main inequality (holds for any n and any text length N multiple of n)

$$\boxed{H(P) \leq \sum_{t=1}^n H(q_t) \leq n H(p)}. \quad (2)$$

Proof. *Step 1 (comparison with product of marginals).* By construction, P has marginals $q_1, \dots, q_n$. Consider the product distribution

$$Q := q_1 \otimes q_2 \otimes \dots \otimes q_n,$$

on $\mathcal{A}^n$. Then

$$H(Q) = \sum_{t=1}^n H(q_t).$$

Since P and Q share the same marginals, one has

$$D(P\|Q) = \sum_w P(w) \log \frac{P(w)}{Q(w)} \geq 0.$$

Rearranging gives

$$H(P) \leq H(Q) = \sum_{t=1}^n H(q_t).$$

Step 2 (concavity + (1)). Entropy is concave, so

$$\frac{1}{n} \sum_{t=1}^n H(q_t) \leq H\left(\frac{1}{n} \sum_{t=1}^n q_t\right) = H(p).$$

Thus $\sum_{t=1}^n H(q_t) \leq n H(p)$. Combining with Step 1 proves (2).

Remarks

- With non-overlapping blocks, each q_t need not equal p ; only the average relation (1) is guaranteed.
- The bound (2) is purely combinatorial: it holds for *every* fixed text, for *every* $n \geq 1$, and for *every* length N that is a multiple of n .